IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

DISTRIBUTED OPTIMISATION AND LEARNING

Monday, 28 April 2025, 10:00 AM

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

This is an OPEN-BOOK examination.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : S. Vlaski
Second Marker(s) : K.K. Leung

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Final Exam (Spring 2025)

ELEC70082 – Distributed Optimisation and Learning

Instructor: Stefan Vlaski

28/04/2024, 10:00 – 13:00

Read all of the following information before starting the exam:

- The exam is open book. Feel free to use any written or printed material you have brought. You can use and refer to any results from the lecture notes that might be useful in answering the questions.
- Calculators or other electronic devices are not allowed for this exam.
- There is a total of 3 problems that total to 100 marks. There is a total of 4 pages.
- You have 3 hours (10:00 - 13:00) to complete the exam.

Question 1. The following questions require true/false answers and justification. Full credit will only be given for correct answers with correct justification, but partial credit is possible.

(a) (true/false) Incremental variants of decentralised algorithms generally exhibit improved performance in terms of their steady-state mean-squared deviation. [5 marks]

(b) (true/false) The bias of penalty-based decentralised optimisation algorithms (such as distributed gradient descent and diffusion) is amplified over sparse network topologies. [5 marks]

(c) (true/false) Suppose we employ exact gradients, and hence the gradient noise is zero. Then, for small step-sizes μ, the difference in steady-state mean-squared deviation between diffusion and Exact diffusion will be negligible. [5 marks]

(d) (true/false) Decentralised algorithms for optimisation and learning offer privacy guarantees by design, because they avoid sharing raw data and instead share models. [5 marks]

(e) (true/false) Consider a collection of $K = 100$ agents. We examine two separate modes of operation. In arrangement (1), the agents are all connected to a fusion-center, and collaborate via centralised stochastic gradient descent with full participation. In arrangement (2), the agents are connected with each other over a sparse, but connected network topology and collaborate via the EXTRA algorithm. The stochastic gradient approximations and all other parameters are identical between the two arrangements. Then, for moderateley large step-sizes (small enough so the algorithms converge, but not so small that higher-order terms can be neglected) and gradient noise levels the two strategies will exhibit approximately the same steady-state mean-squared deviation. [5 marks]

(f) (true/false) Given two primitive matrices A_1 and A_2, we define the matrix: [5 marks]

$$A \triangleq \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}$$

Then A is also primitive.

(g) (true/false) The rate of convergence of the diffusion algorithm is unaffected by the mixing matrix of the combination matrix, provided that the step-size is sufficiently small. [5 marks]

(h) (true/false) Consider a star-graph, where agent 1 is connected to every other agent (and itself), and every other agent is only connected to agent one (and itself). Then, the entry of the Perron vector corresponding to agent 1 will be larger than all other entries. [5 marks]

Question 2. We consider a collection of K agents, where each agent k is equipped with a local mean square error risk:

$$J_k(w) = \frac{1}{2}\mathbb{E}\Big(\boldsymbol{\gamma}_k - \boldsymbol{h}_k^{\mathsf{T}} w\Big)^2$$

The data follows the local linear models:

$$\boldsymbol{\gamma}_k = \boldsymbol{h}_k^{\mathsf{T}} w^o + \boldsymbol{v}_k$$

Note that the regressors $\boldsymbol{\gamma}_k$ and noise $\boldsymbol{v}_k$ are distinct across agents, but the linear model w^o is identical. We are interested in solving the consensus optimization problem:

$$w^o = \arg\min_{w\in\mathbb{R}^M} \frac{1}{K}\sum_{k=1}^{K} J_k(w)$$

(a) Suppose each agent k observes a single pair $\{\boldsymbol{h}_{k,i}, \boldsymbol{\gamma}_{k,i}\}$ at time i. All observations (across space and time) are independent. Formulate the recursions for the diffusion and for the NEXT algorithm when agents perform local elementary gradient approximations. [10 marks]

(b) Suppose agents only observe a pair $\{\boldsymbol{h}_{k,i}, \boldsymbol{\gamma}_{k,i}\}$ with probability π_k, and no data otherwise. How would you need to adjust the recursions from part (a) to ensure that you continue to have a working algorithm with unbiased local gradient approximations? [10 marks]

(c) Determine expressions for the steady-state mean-squared deviation of both the diffusion and NEXT implementations from part (b). [10 marks]

Question 3. We consider an incremental variant of the exact consensus algorithm for tracking the average of a collection of signals $\{\boldsymbol{g}_{k,i}\}_{k=1}^{K}$ over a network:

$$\begin{aligned}\boldsymbol{\psi}_{k,i} &= \boldsymbol{w}_{k,i-1} + \boldsymbol{g}_{k,i} - \boldsymbol{g}_{k,i-1} \\ \boldsymbol{w}_{k,i} &= \sum_{\ell \in \mathcal{N}_k} a_{\ell k} \boldsymbol{\psi}_{\ell,i}\end{aligned}$$

The signals $\boldsymbol{g}_{k,i} \in \mathbb{R}$ are scalars, and we initialize $\boldsymbol{w}_{k,0} = \boldsymbol{g}_{k,0}$. The combination weights are collected in a primitive matrix A that satisfies:

$$\begin{aligned}A &= A^{\mathsf{T}} \\ A\mathbb{1} &= \mathbb{1}\end{aligned}$$

We model the signals as follows:

$$\boldsymbol{g}_{k,i} = \begin{cases}\boldsymbol{g}_{k,i-1}, & \text{with probability } 1-\epsilon, \\ \boldsymbol{g}_{k,i-1} + \boldsymbol{r}_{k,i}, & \text{otherwise.}\end{cases}$$

where $\boldsymbol{r}_{k,i} \sim \mathcal{N}(0, \sigma_r^2)$. This means that at any time i, the signal $\boldsymbol{g}_{k,i}$ will remain at $\boldsymbol{g}_{k,i-1}$ with probability $1-\epsilon$. With probability ϵ, it will move by a random value $\boldsymbol{r}_{k,i}$, which is modelled as a normally distributed random variable with common variance σ_r^2. This means that ϵ close to 0 indicates a stable environment, while ϵ close to 1 indicates a highly volatile environment.

(a) Determine an expression for the mean and variance of $\boldsymbol{g}_{k,i} - \boldsymbol{g}_{k,i-1}$. [10 marks]

(b) Determine an expression for the network centroid: [10 marks]

$$\boldsymbol{w}_{c,i} \triangleq \frac{1}{K} \sum_{k=1}^{K} \boldsymbol{w}_{k,i}$$

in terms of the signals $\boldsymbol{g}_{k,i}$.

(c) Determine an expression for the rate of convergence of the tracking error [10 marks]

$$\sum_{k=1}^{K} \mathbb{E}\|\boldsymbol{w}_{k,i} - \boldsymbol{g}_{c,i}\|^2,$$

relative to the signal centroid $\boldsymbol{g}_{c,i} = \frac{1}{K}\sum_{k=1}^{K} \boldsymbol{g}_{k,i}$, as well as its limiting value:

$$\limsup_{i \to \infty} \sum_{k=1}^{K} \mathbb{E}\|\boldsymbol{w}_{k,i} - \boldsymbol{g}_{c,i}\|^2$$